

Sequents: An Ordinal Theory for Solana Tokens

Abstract. A fungible token is designed to forget. Once two units sit in the same account nothing tells them apart, and the record of how each one arrived is gone. Collectors of physical money have always cared about that record, and nonfungible tokens later showed how strongly people respond to rarity and to the moment it is revealed, though they did it by making every item a separate asset. Ordinal theory found another way on Bitcoin: number the smallest units in the order they were created, follow them through transactions with a published rule, and let rarity come from history without changing the asset or the chain. That rule depends on the shape of Bitcoin transactions and does not carry over to Solana, where token balances merge inside accounts. We propose Sequents, a set of rules under which every unit of a Solana token receives a position from the order in which it was first bought. Positions are grouped into Strikes, rarity survives only where its history can be proven, and rare units change hands inside sealed envelopes without ever moving. Traits are assigned from a public seed that nobody can know while the units are being sold, and every result can be recomputed by anyone from the public ledger.

1 Introduction

Money is useful because its units are interchangeable. A merchant who accepts a quarter does not ask which quarter it is. Yet collectors have asked that question for as long as coins have existed. A United States quarter struck in 1964 contains silver and one struck a year later does not. A coin pressed twice by a faulty die can sell for thousands of times its face value. At the shop both are worth twenty five cents, and at a dealer's table they are worth very different amounts. The asset is the same in both places. What differs is whether anyone is reading its history.

Digital tokens on Solana are fungible in a stronger sense than coins. A coin at least keeps its date and its flaws in the metal. A token keeps nothing: its account stores a single number, and when tokens from two sources arrive in the same account they become indistinguishable. There is no date to read.

This paper began with a casual question asked in public: could one holder's share of a memecoin be rarer than another's, without turning the coin into an NFT? The replies noticed the difficulty at once. If some units are rarer than others then the token sounds nonfungible, and if it is nonfungible it sounds like an NFT. One reply pointed out that Ordinals had already attempted something similar on Bitcoin with rare satoshis.

Our answer is that rarity does not need to live inside the asset. It can live in a reading of the asset's history, provided the rules of that reading are public, fixed in advance and impossible to bend afterwards. The token stays one ordinary fungible token, priced identically everywhere it trades. On top of it we define a convention that gives particular units a past, and we show how that convention can be made precise on a chain whose design works against it.

2 Ordinal Theory

Casey Rodarmor described ordinal numbers for Bitcoin in a draft proposal posted to the Bitcoin developers' mailing list in February 2022 [2]. The idea is simple to state. Every satoshi is numbered in the order it was mined, so the first satoshi of the first block is number zero and the count continues through every block reward. When a transaction spends inputs and creates outputs, the satoshis of the inputs are lined up in order and poured into the outputs in order, first in, first out. Because a Bitcoin transaction lists its inputs and outputs in a fixed sequence, this rule tells you exactly which numbered satoshis ended up where, forever.

An indexer applying the rule does not need to track two quadrillion individual numbers. It tracks contiguous ranges, which split only when a transaction divides them. Rodarmor noted that a variation of the scheme had been proposed on BitcoinTalk almost a decade earlier [3], which says something about how natural the idea is once the transaction structure is understood.

Rarity follows from Bitcoin’s own calendar. The first satoshi of each block is called uncommon, the first of each difficulty adjustment period rare, the first of each halving epoch epic, the first of each cycle legendary, and the first satoshi of the genesis block mythic [4]. Every one of these is objective and known the instant the satoshi is mined. Collectors have since added historical categories, such as satoxis from block 9, which apply to whole blocks rather than single units.

The lasting lesson of ordinal theory is that none of this required a change to Bitcoin. Nodes do not know that ordinal numbers exist. The numbering is a convention, the software that applies it is open, and anyone who runs it reaches the same answer. A rare satoshi is still worth one satoshi to a merchant. Its premium exists only where collectors meet to trade it.

3 Related Work

3.1 Nonfungible tokens

The nonfungible token standard on Ethereum [5] made each item in a collection a distinct asset with its own identifier and metadata. Within a few years these collections had developed conventions that matter to this paper.

The first is the reveal. Many collections were sold blind: buyers paid for an item before knowing its traits, and the traits were disclosed together at a later moment. That moment became a cultural event in its own right. The second is the provenance hash. Careful projects, the Bored Ape Yacht Club among them, published a cryptographic hash of their final images and metadata before the sale began, so that buyers could later confirm nothing had been rearranged after the money arrived [6]. The third is rarity ranking. Tools such as rarity.tools ranked items by how uncommon their combination of traits was, and these rankings shaped prices [7]. The fourth is collection size. Collections settled on numbers that people could browse and remember, often ten thousand.

The failures were as instructive as the successes. In many sales the metadata could be read before the official reveal, and people who knew where to look bought the rare items first. In others the team itself knew the traits in advance, and buyers had no way to prove that this knowledge had not been used. Randomness was sometimes weak. In 2021 the Meebits sale by Larva Labs was exploited by a buyer who repeated attempts and abandoned the unfavourable ones until a mint produced one of the rarest traits [8]. And a great deal of metadata lived on ordinary web servers, where it could change or disappear.

Sequents takes the reveal, the commitment and the idea of a browsable collection from this history. It takes its fairness rules from the failures.

3.2 Hybrid tokens

Experimental standards such as ERC404 on Ethereum [9] and MPL404 on Solana [10] link a fungible token to a set of nonfungible tokens, so that holding enough of the fungible token corresponds to holding an item, and the two can be exchanged. In some designs the item’s traits are rerolled on each exchange. These systems

answer a different question. Their rarity is assigned metadata attached to a separate asset, and it can change. In Sequents there is no second asset, and rarity is not assigned to a holder at all. It is read from the history of the units themselves, and that history cannot be rerolled.

3.3 Bitcoin meta protocols

BRC20 in 2023 and Runes in 2024 extended the ordinal idea to fungible balances defined entirely by indexer rules applied to Bitcoin transactions. They are further evidence that a published convention, read consistently by open software, can carry real economic meaning without any change to the underlying chain.

4 The Problem on Solana

Ordinal theory works because a Bitcoin transaction consumes specific outputs and creates specific outputs in a defined order. The structure of the transaction is itself the answer to the question of which satoshis went where.

Solana has no such structure for tokens. A token account stores one integer, its balance. A transfer instruction moves an amount from one account to another and says nothing about which units. Tokens that arrive from different places merge into the same integer. Swaps route through pools and intermediate accounts owned by programs, often several in a single transaction, and a pool may hold tokens deposited by thousands of people.

One could try to invent an ordering for every account and every transfer and follow units everywhere. We considered this and rejected it. Once units enter a shared pool, any rule for deciding which ones come out is arbitrary, and any arbitrary rule that can be computed in advance can be gamed by whoever computes it fastest. Following units through every program on the network would also mean that every edge case, in every program, becomes part of the protocol.

The design that follows takes the opposite approach. It follows units only where their history can be stated without invention, and it says plainly that everywhere else the history ends.

5 Positions

A token launched on pump.fun is first sold from a bonding curve: a program that holds the saleable supply in a single account and sells it at a price that rises with each purchase. While the curve is active it is the only source of new units, and every purchase from it is recorded on Solana in a total order, by slot and by position within the block.¹

This gives us something Bitcoin gives ordinal theory for free: a place where units are created in sequence. We number them there.

The saleable supply on the standard curve is $S = 793,100,000$ tokens, each divisible into 10^6 base units. Every saleable base unit receives a *position* p with $0 \leq p < S \cdot 10^6$. A curve purchase of n fresh units is issued the positions $[c, c + n)$, where c is the number of positions already issued, and the counter advances by n . Positions are issued exactly once. If a holder sells units back to the curve, those units become ordinary, and a later buyer at that depth receives ordinary units rather than reissued positions. The roughly 206.9 million

¹That order is established by Solana's leader schedule and its Proof of History clock [11]. We rely only on the fact that it is total and public.

tokens that never pass through the sale, and which move to a trading pool when the curve completes, are ordinary from the start.

Positions are written in whole tokens as a Strike number and an offset, so that the token at position $7,412,000 \cdot 10^6$ is written 7.412000. The analogy with ordinal numbers is direct. Mining order becomes purchase order, and the satoshi's number becomes the token's position.²

6 Strikes

Positions are grouped into *Strikes* of one million tokens each. Strike s contains the positions

$$s \cdot 10^{12} \leq p < \min((s+1) \cdot 10^{12}, S \cdot 10^6),$$

so there are 794 Strikes, numbered 0 to 793, and the last holds 100,000 tokens. In coinage a strike is a single coin pressed from a die. Here it is a block of a million tokens pressed by the curve, one after another.

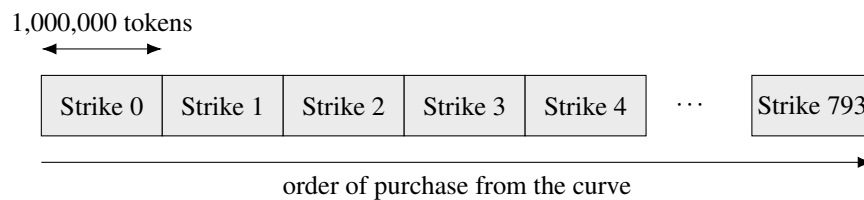


Figure 1: Positions are issued in purchase order and grouped into 794 Strikes.

Traits belong to Strikes rather than to individual purchases. If traits were drawn once for each purchase, a single large buy covering thirty million tokens would receive one draw and so would a buy of fifty thousand tokens, which is unfair to both. With Strikes, every million tokens receives exactly one chance, however they were bought. In ordinal terms, Strikes resemble blocks, and their traits resemble the historical categories that apply to every satoshi in a block.

The size of a Strike is a deliberate choice. One Strike is a tenth of a percent of total supply, roughly the size of a small holder's position, so ordinary buyers can own whole Strikes. At the same time 794 is a collection small enough to browse, discuss and remember, which is the same reasoning that led nonfungible collections to settle on sizes people could hold in their heads.

7 Survival

Rarity in Sequents can exist in exactly two places. The first is the *origin account*, meaning the token account that received the positions in a curve purchase. The second is an *envelope*, described in the next section. Tokens anywhere else are ordinary.

When tokens leave an origin account by any route, whether a swap, a transfer to another person, a sale back to the curve or a deposit into a pool, the rare positions among them *melt*. They become ordinary units permanently, and the destination receives ordinary units. If the ownership of the origin account itself changes,

²The name of the protocol comes from this. A *sequent* is something that follows in order, from the Latin *sequi*. In logic, a sequent is the basic statement of Gentzen's sequent calculus [12], one of the foundations of proof theory. Ordinal numbers belong to set theory. It seemed fitting that a protocol whose first token is called \$PROOF should take its name from proof theory instead.

every rare position in it melts. Tokens that arrive in an origin account from anywhere other than the curve are ordinary even there.

When only part of a balance leaves, the rule must say which units go. Ordinary units leave first. After that, rare positions leave in order of ascending rank, so the least rare go before the most rare, and among positions of equal rank the highest positions go first. A holder can therefore sell freely up to the size of their ordinary balance without touching anything rare.

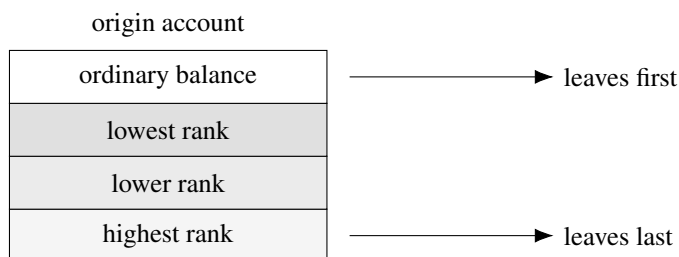


Figure 2: The order in which tokens leave an origin account. Rare positions that leave become ordinary.

This rule is the answer to the problem of Section 4. It never asks which units emerged from a pool, because nothing that enters a pool is rare. The indexer needs to watch only three things: the curve, the envelope program and the accounts that currently hold rare positions. It never has to interpret an arbitrary program's internal movements.

The rule has a consequence that we regard as its most important property. Rare supply can only fall. Positions are issued once, melted positions never return, and no event after the commitment can create rarity. If $R(t)$ is the number of rare positions surviving at time t after the sale is complete, then R is nonincreasing, and every melt makes the survivors scarcer.

Ordinal theory makes a different choice. A rare satoshi stays rare wherever it goes, even into a transaction fee. The closer physical analogy for Sequents is melting a silver coin for its metal. The metal still has value, but the coin is gone as a collectible, and every coin melted makes the remaining ones scarcer.

8 Envelopes

Survival alone would leave rare positions with nowhere to go. They could be held, but any attempt to pass them on would melt them. Envelopes solve this.

An envelope is an account created by the Sequents vault program at a program derived address, which means no private key for it exists. Each envelope owns a token account, the vault, whose authority is the envelope itself. The envelope records a holder, a status, a price and the positions it was sealed with.

A holder creates an envelope by *sealing*: they name up to eight ranges of positions in their origin account, and the program moves exactly that many tokens into a new vault. This is the only time sealed tokens ever move. The indexer confirms that the named ranges really were rare in that account. If they were not, the envelope holds ordinary tokens, and it is shown that way.

After sealing, an envelope can be listed at a price, withdrawn from sale, sold or given away. A sale is a single transaction in which the buyer's payment goes to the holder and the envelope's holder record becomes the buyer. The tokens do not move. The buyer can set a maximum price, so a seller who raises the price while the transaction is in flight causes it to fail rather than overcharge. The only way tokens ever leave a vault is

withdrawal, which returns them to the holder as ordinary tokens and closes the envelope. Withdrawal melts.

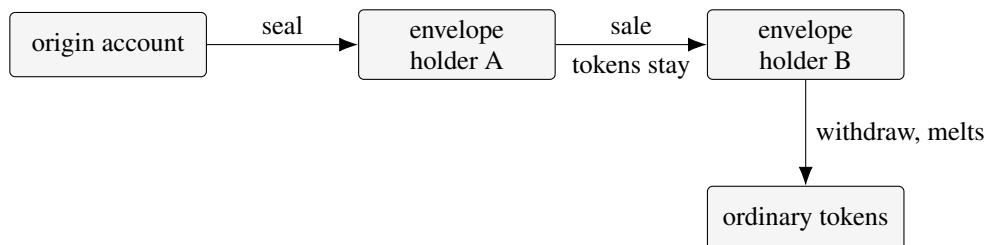


Figure 3: Tokens move once when sealed. Sales change the holder record. Withdrawal is the only exit.

This divides the guarantees cleanly. The program guarantees custody: who controls an envelope, and that tokens leave a vault only one way. The indexer guarantees identity: which positions are inside, by the published rules. The program cannot see positions and does not need to. The indexer cannot move tokens and does not need to.

Ordinal marketplaces reach a similar result differently. A Bitcoin seller can sign a partial transaction that spends a specific output only if the buyer adds the agreed payment, so the satoshis wait in the seller’s own wallet until the sale. Solana transactions expire within about a minute of being signed, which makes that pattern awkward. A program account that holds the tokens and records the holder achieves the same effect with less fragility.

9 Traits and the Reveal

9.1 Traits

Every Strike has a date trait and may have one error. Date traits are positional and visible from the moment a Strike is sold. Errors are random, named after real minting errors, and assigned at the reveal.

Kind	Trait	Strikes	Points
Date	Genesis	Strikes 0 to 4	40
Date	Key Date	Strikes 5 to 24	15
Date	Final Strike	Strike 793	10
Date	Common Date	all others	0
Error	Double Die	3	100
Error	Wrong Planchet	6	70
Error	Off Center	12	50
Error	Clipped Planchet	24	30
Error	Die Crack	48	15

Table 1: Traits of the first Sequents token. Ninety three Strikes, about 11.7 percent, receive an error.

9.2 Rank

A Strike’s rank is its date points plus its error points. Higher ranks are rarer, and rank determines the order in which rare positions leave an account. Before the reveal every rank is zero.

Nonfungible collections usually rank items by statistical rarity, multiplying the frequencies of their traits. When we tested that method on this collection it produced orderings nobody would defend, with a plain Strike carrying a scarce positional trait ranking above a much scarcer error. Explicit points state the intended hierarchy directly. Errors dominate, early position adds weight, and a late Strike with a Double Die outranks any plain Genesis Strike. For display, ranks are grouped into four bands named after ordinal theory's scale: Common, Uncommon, Rare and Legendary.

9.3 The reveal

The reveal is designed around the failures described in Section 3.1. It proceeds as follows.

Before the first unit is sold, the rules file, which fixes every trait, count and point value, is committed to Solana in a signed memo together with a deadline slot. The memo carries the SHA256 hash of the rules file. A commitment made after the first sale is invalid, and without one there is no rarity at all.

When the curve sells its last position, or at the deadline if that comes first, a target slot is fixed: $T = \min(\text{completion} + 150, \text{deadline})$. About a minute after the sale ends, in other words. The Strikes eligible for errors are those fully sold by T . The seed block is the first block produced at or after T , and the seed is the SHA256 hash of a fixed label followed by that block's hash. From the seed, a counter driven stream of SHA256 outputs produces uniform random integers, and those integers shuffle the eligible Strikes by the Fisher and Yates method. The first three Strikes in the shuffled order receive Double Die, the next six Wrong Planchet, and so on down the table. Finally, a reveal file containing the rules, the slots, the block hash and every derived trait is published, and its hash is committed in a second memo.

Each step answers a specific failure. The commitment is a provenance hash, improved in one respect: it commits to rules rather than to finished results, so nobody, including the authors, ever holds a secret list of which Strikes are rare. No one can read traits early, because they do not exist until a block that is produced after the sale has ended. No one can retry their way to a rare Strike, because Strikes still unsold at T cannot receive errors, and units bought after the reveal from a Strike already known to be rare are ordinary by the survival rule. And nothing that matters is stored on a server: the rules and the seed are on Solana, and the derivation is open code.

10 Verification

Nothing in Sequents asks to be trusted. Anyone can carry out the following steps independently of the authors.

To verify the traits, read the commitment memo and confirm that the published rules file has the committed hash. Determine the completion slot and the target slot from the public ledger, find the first block produced at or after it, and take its hash. Run the derivation. The same code runs in the indexer and in an ordinary web browser, and assigning traits to 794 Strikes takes a fraction of a second. Compare the result with the reveal file whose hash is in the second memo. Any disagreement is visible to everyone.

To verify holdings, run the open indexer. It reads only finalized, successful transactions, in the order Solana established. At every step it checks the balance it has computed for each account it tracks against the balance Solana itself reports, and if they ever differ it stops rather than guess. At regular intervals it publishes a fingerprint, the SHA256 hash of its entire ledger in a canonical form. Two indexers run on independent infrastructure, reading from independent providers, and their fingerprints must agree.

To verify an envelope before buying it, read its account directly from Solana: the program, the mint, the holder, the status, the price and the vault balance. Then confirm with the indexer that the positions inside are the ones the listing claims.

This is the same arrangement that gives ordinal theory its credibility. The chain records what happened. The rules say what it means. The software that applies the rules is open, and it is deterministic, so honest implementations always agree.

11 Calculations

Suppose N Strikes are eligible for errors, which is 794 if the curve completes, and an error is assigned to c of them uniformly at random. A holder who owns k whole Strikes receives that error at least once with probability

$$P(k, c) = 1 - \frac{\binom{N-k}{c}}{\binom{N}{c}},$$

and the expected number of Strikes they hold carrying it is kc/N . Across all errors, $c = 93$.

Share of supply	Strikes owned	Double Die	Wrong Planchet	Off Center	Clipped Planchet	Die Crack	Any error	Expected errors
0.1%	1	0.4%	0.8%	1.5%	3.0%	6.0%	11.7%	0.1
0.5%	5	1.9%	3.7%	7.3%	14.3%	26.8%	46.5%	0.6
1%	10	3.7%	7.3%	14.2%	26.6%	46.6%	71.4%	1.2
3%	30	10.9%	20.7%	37.2%	60.9%	85.1%	97.8%	3.5
10%	80	27.3%	47.2%	72.3%	92.5%	99.5%	100%	9.4

Table 2: Probability of holding at least one Strike with each error, with $N = 794$.

The table shows the balance the error counts were chosen to strike. A holder of a single Strike has roughly a one in nine chance of holding some error. A holder of three percent of supply will almost certainly hold one, yet has only about a one in nine chance at a Double Die. Random traits are shared in proportion to ownership. Nobody can buy all of the rarest traits without buying most of the supply.

Positional traits behave differently. Genesis is the first five million tokens, and on the standard curve those cost roughly 0.14 SOL in total. A single early purchase can hold all of Genesis. This is why date points are deliberately small compared with error points.

Survival adds a second effect over time. A Strike’s *intact fraction* is the share of its positions that are still rare. It starts at one when the Strike is sold and can only fall. The scarcity of a trait is therefore not fixed at the reveal. It is the number of Strikes carrying the trait, weighted by how much of each survives, and it moves in one direction only.

12 Limitations

Rarity in Sequents is a social convention. Solana does not enforce it, wallets do not display it and exchanges do not price it. Every unit of the token trades at the same price on every public market. Like any collectible, and like nonfungible tokens before it, a premium exists only to the extent that collectors choose to pay one, and nothing in this paper suggests that they will.

Early positions concentrate. Whoever buys first from the curve holds the first positions, and on pump.fun nothing limits the size of an early purchase. The design responds by keeping positional traits small and by making most of the value random, but it does not remove the advantage of being early.

The producer of the seed block has some influence over that block's hash, and the schedule of block producers is public in advance. Steering the hash toward a favourable outcome would require that producer to care about this token and to act within a fraction of a second. We consider the risk small. A verifiable random function would remove it at the cost of an outside dependency.

Envelopes are only as safe as the vault program that holds them. Its upgrade authority is held by a public multisignature account with a published delay until an external audit, after which it is removed and the program becomes immutable.

The constants in this paper reflect the standard pump.fun curve at the time of writing. The rules file committed at launch is authoritative, and the values in it should be checked against the live token.

13 Conclusion

We have described a way to give the units of a fungible Solana token a history without making them anything other than fungible. Units are numbered in the order they leave the curve and grouped into Strikes. Rarity survives only where its history can be proven, and it can only decrease. Rare units change hands inside envelopes that never move them. Traits come from a seed that does not exist while the units are being sold, and every step can be checked by anyone with access to the public ledger. Every token costs the same. Some of them are rare.

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